

Review Article

A systematic review of artificial neural network techniques for analysis of foot plantar pressure

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ABSTRACT

Plantar pressure distribution offers insights into foot function, gait mechanics, and foot-related issues. This systematic review presents an analysis of the use of artificial neural network techniques in the context of plantar pressure analysis. 60 studies were included in the review. Sample size, pathology, pressure sensor number, data collection device, utilization of other sensor devices, ground-truth methods, pre-processing dataset, neural network type, and evaluation metrics were evaluated. Utilization of customized wearable footwear devices for the acquisition of data was common amongst both healthy participants and patients. Inertial measurement units emerged as an effective compensatory measure to address the limitations associated with the distribution of plantar pressure. Ground truth methods predominantly relied on the usage of both annotations and reference devices. Multilayer perceptron, convolutional neural networks, and recurrent neural networks were identified as the most frequently employed artificial neural network algorithms across the reviewed studies. Finally, the evaluation of performance largely drew upon statistical descriptions and other machine learning methods. This review provides a comprehensive understanding of the use of artificial neural network techniques in plantar pressure analysis, highlighting opportunities for future research.

1. Introduction

Plantar pressure distribution describes the magnitude and spread of force or load across the plantar surface of the foot during standing, walking, and other weight-bearing activities. The analysis of plantar pressure provides valuable insights about foot function [1], gait mechanics [2], and the development of foot-related pathologies [3]. Understanding plantar pressure distribution during functional activities can help healthcare professionals, researchers, and footwear designers develop effective prevention [4,5] and management strategies [6–8] and promote optimal foot health [9] and performance [10].

The relevance of this study stems from the growing recognition of artificial neural network (ANN) techniques as powerful tools in healthcare and biomechanics. Unlike traditional analytical methods, ANNs can handle complex, high dimensions, nonlinear data efficiently, offering more nuanced and accurate insights into foot mechanics [11]. This is

particularly valuable in foot plantar pressure analysis, where the intricacies of pressure distribution patterns are crucial for understanding various pathologies [12] and designing effective interventions [13].

Both Razak et al. [1] and Chen et al. [2] conducted reviews on foot plantar sensors and pressure measurement systems, underscoring the significance and practicality of employing machine learning to elucidate the intricate relationships between foot plantar pressure and human lower limb locomotion. However, these reviews did not encompass a thorough analysis of the application of ANNs in the context of foot plantar pressure. In contrast, Mei et al. [14] presents a review focusing on the utilization of machine learning and wearable sensors to predict freezing of gait in Parkinson's patients. While this study offers comparisons of sensors and machine learning algorithms, it falls short in addressing the implications of multisensory fusion conditions and lacks a broader applicability to general human lower limb locomotion scenarios. Tulloch et al. [15] systematically reviews the application of

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machine learning in preventing, diagnosing, and managing diabetic foot ulcers (DFUs). Notably, the majority of the articles reviewed in this study predominantly employ non-Artificial Neural Network (NANN) machine learning techniques.

Despite advancements in technology and methodology, there remains a significant gap in the application and understanding of ANNs in the analysis of foot plantar pressure. This systematic review addresses this gap by providing an in-depth analysis of ANN techniques in the context of foot plantar pressure, a topic of increasing importance for both clinicians, biomechanical researchers, and footwear device industry.

1.1. Measurements

Plantar pressure is commonly assessed using devices such as pressure platforms or in-shoe pressure sensors. Pressure platforms, which can be integrated into the floor, contain sensors that measure the magnitude and timing of forces exerted during standing, walking, or running [16]. Whilst providing comprehensive information about plantar pressure, disadvantages of pressure platforms include the spatial measurement area and lack of real-world environmental factors. Plantar pressure platforms can also be expensive to purchase, install, and maintain. These factors limit accessibility for some research or clinical settings, making it challenging to conduct large-scale studies or use them routinely in certain environments. Conversely, in-shoe pressure sensors are thin flexible devices placed inside a person’s shoes which allows plantar pressure to be measured for longer periods and during activities of daily living [17]. However, concerns regarding the level of detail, accuracy [1], placement and fit [2], influence of footwear [18], cost and technical expertise [19] have been raised.

Common methods of analysing plantar pressure involve: 1) static pressure distribution maps that provide a visual representation of pressure patterns across the plantar surface [20]; 2) vertical ground

reaction force (GRF) that refers to the total force exerted by the ground on the body during walking or running [21]; 3) peak pressure and pressure–time–integral that provide pressure exerted at a specific location on the foot during a gait cycle [22]; 4) centre of pressure (COP) that provides information about foot stability, balance, and weight distribution [23].

However, analysis of plantar pressure typically provides two-dimensional rather than detailed three-dimensional information. Two dimensional information, such as movement and shear forces, is important for understanding foot function and biomechanics [24]. Two-dimensional plantar pressure analysis also lacks contextual information such as joint angles and muscle activation patterns i.e., kinetic, and kinematic data [25]. Integrating plantar pressure analysis with other measurement techniques, such as motion analysis or electromyography (EMG), can provide a more holistic understanding of foot function during normal daily activities [26].

It is possible, and increasingly common, to utilize multidimensional information from sensors, such as inertial sensors (e.g., accelerometers, gyroscopes) [27], EMGs [28], and RGB-D cameras [29], in conjunction with plantar pressure sensors for analysis. Sensor fusion methods can be employed to integrate data from multiple sensors and extract more comprehensive insights of foot biomechanics. As proposed by Ramachandram and Taylor [30], the fusion of multisensory data can be segregated as data-level, feature-level, and decision-level fusion to set connection among different sensor data (Fig. 1). However, the feature extractions and alignment can be computationally complex and almost impossible for human beings. As a result, it is necessary to utilize ANNs to assist data analysis.

1.2. Machine learnings for plantar pressure and gait analysis

Measuring plantar pressure can be time-consuming, even for experienced healthcare professionals and researchers [31]. Additionally,

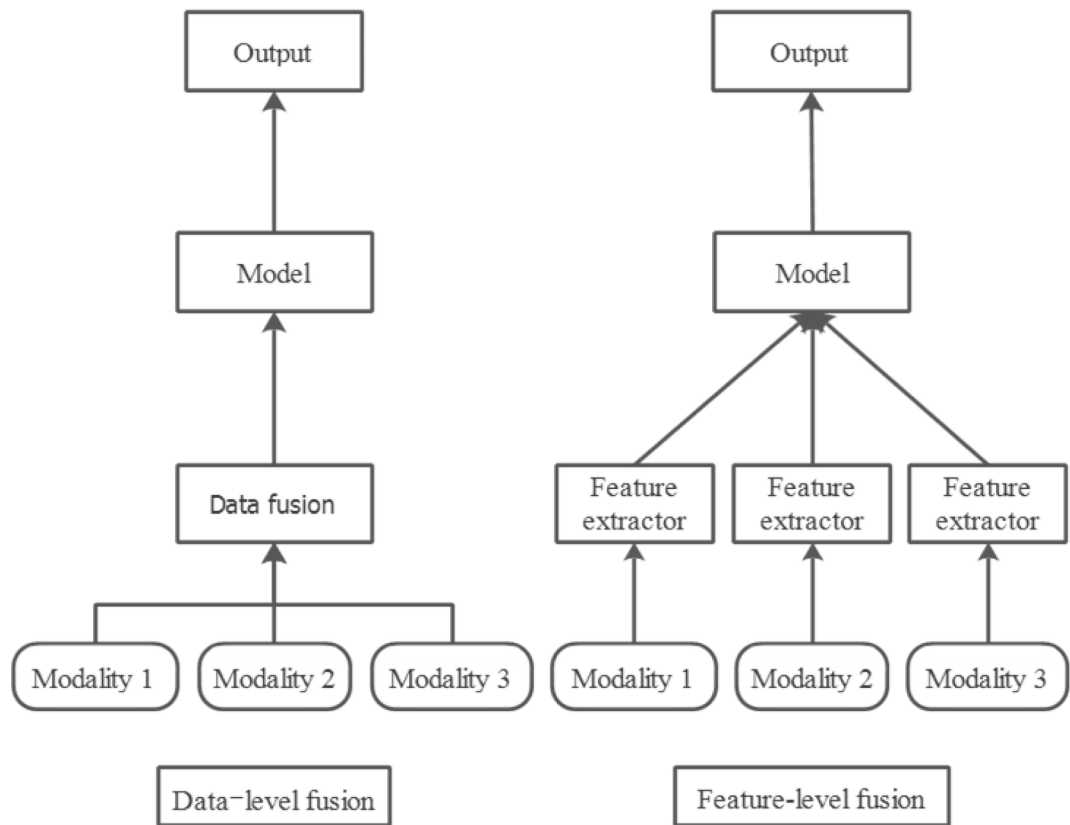


Fig. 1. Data-level and feature-level fusion.

advancements in technology, in terms of equipment and the amount and type of information that can be collected, have meant that it is almost impossible to apply all available techniques simultaneously, particularly in clinical settings. As a result, there has been increased interest in the application of machine learning algorithms for feature extraction [32], image segmentation [33], complex pattern recognition [16], personalized interventions [34], image super-resolution [35] and data reconstruction [36] etc. Table 1 lists popular machine learning algorithms for plantar pressure analysis.

While NANN methods, such as traditional machine learning analyses, have been instrumental in the past, they often fall short in dealing with the high-dimensional and complex data typical in plantar pressure studies, especially when modulated with other sensors. ANNs, on the other hand, excel in extracting meaningful spatial–temporal patterns from high-dimensional data, making them increasingly preferred in advanced biomechanical research [11].

1.3. Main contribution

Our study primarily targets biomechanics researchers, clinical practitioners, and data scientists keen on understanding the intersection of ANNs and foot plantar pressure analysis. Through thorough article selection and quality assessment, this review critically selected 60 articles in the combined field of ANNs and plantar pressure distribution analysis, categorizing them based on population, data collection methods, multisensory fusion, ground truth, ANN algorithms, and evaluation techniques. By providing a comprehensive analysis of these

Table 1
Popular machine learning techniques.

Technique	Description
NANN	Simple algorithms, often employed for classification or regression tasks. Examples include random forest (RF) [37], Naive Bayes (NB) [32], support vector machines (SVM) [38], k-nearest neighbours (KNN) [39], and fuzzy logic [40].
Feedforward Neural Networks (FNNs)	The basic types of ANN where information flows in one direction, often used for classification, regression, and pattern recognition tasks [36,37]. Multi-layer perceptron (MLP) is a commonly used FNN variant [6].
Convolutional Neural Networks (CNNs)	Deep learning models of FNN designed for analysing visual data, capturing local and global patterns in images. CNNs have revolutionized computer vision tasks and achieved state-of-the-art results [43].
Recurrent Neural Networks (RNNs)	Feed backward neural network architecture specialized in handling sequential data, leveraging feedback connections to retain information from previous time steps [42]. Variants like long short-term memory (LSTM) [44] and gated recurrent units (GRUs) [45] improve upon some limitations of traditional RNNs.
Generative Adversarial Networks (GANs)	Deep learning models consisting of a generator and discriminator networks. GANs generate synthetic data resembling real data by training the generator to produce realistic samples and the discriminator to distinguish between real and generated samples [46].
Autoencoders	Unsupervised learning models used for feature extraction and dimensionality reduction. Autoencoders learn a compressed representation of input data and can reconstruct the original data from this representation. Autoencoders are useful for learning meaningful representations and various applications in machine learning and data analysis [47].
Transformers	The transformer technique is a deep learning model proposed by Google in 2017 mainly used to handle natural language processing tasks. The transformer technique is able to handle long sequential data, has high parallelism and efficiency, and is becoming increasingly popular in computer vision, speech recognition, and creative arts [48].

included articles, we aim to bridge the gap between clinical implications and computational advancements in this field.

The objective of this systematic review is to critically evaluate the effectiveness of ANN algorithms in plantar pressure distribution analysis and investigate the validation methods employed in relevant studies. Specifically, the aims of this review are to:

- 1) Identify the ANN algorithms used in plantar pressure distribution analysis, categorizing them based on their application for image processing, time sequence analysis, or fusion with other sensors.
- 2) Examine the validation methods employed in studies utilizing ANN algorithms for plantar pressure analysis, with a focus on the use of ground truth data and annotation methods.
- 3) Summarize the evaluation methods and the performance scores of ANN algorithms.
- 4) Identify gaps and challenges in the literature and provide recommendations for future research directions and improvements in the field.

2. Review methodology

2.1. Search strategy and selection criteria

This systematic literature review followed the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) protocol to ensure the study’s replicability [49]. A literature search was conducted in the Web of Science core collection databases in May 2023, encompassing articles published from 2015 to 2023. The literature search strategy employed a combination of keywords to identify articles that addressed plantar pressure distribution, ANNs and sensor fusion. Studies had to be published in English. References were imported into Rayyan [50] and duplicates were removed. C.W. and H.L., both researchers, conducted independent screenings of the titles and abstracts to identify potential articles for inclusion. If the initial screening suggested that a particular study conformed to the predefined criteria, or if the title and abstract did not provide sufficient information for a decision, a thorough review of the full-text article was undertaken. In instances where C.W. and H.L. encountered disagreements, they sought to resolve them through deliberation. If a consensus remained elusive, the intervention of a third researcher, L.Z., was solicited to arbitrate the matter.

2.2. Eligibility criteria

Studies were considered for inclusion if:

- 1) The study focused on plantar pressure, plantar pressure distribution, GRF, or plantar tissue stress.
- 2) Data collection device used at least four force sensors [2] (both force plates and wearable devices were eligible).
- 3) The analysis algorithms utilized ANN techniques (any variant of deep learning techniques such as CNN, RNN, and LSTM. In addition, shallow neural networks such as MLP, FNN, and backpropagation neural network were included).
- 4) The method of ground truth data was provided to work as a benchmark to evaluate the performance of a model.
- 5) At least one statistical analysis (e.g., mean square error, F1 score, or an evaluation metric such as accuracy, correlation, precision, or sensitivity) was used to evaluate the performance of algorithms.

Studies were excluded if:

The research was outside of the scope of the current review e.g., studies which included pedestrian dead reckoning, navigation, exoskeleton or focused on the design of advanced sensor technologies.

Methodologies were confined solely to NANN machine learning techniques or traditional statistical analysis. This exclusion criterion was based on the comparatively limited efficacy of these methods in the analysis of high-dimensional data, as well as their diminishing prevalence in contemporary research paradigms. The study was conducted on animals.

2.3. Quality assessment

To ascertain the quality of the included publications, we utilized an adapted quality checklist based on the critical appraisal skills programme [51] and summarized five systematic reviews in machine learning field [52–56]. The questionnaire was structured to align with machine learning experiment phases: 1) problem definition and background, 2) data collection and preprocessing, 3) model design and selection, 4) model training, 5) model evaluation and validation, 6) results and discussion), and presented in Table 2. Studies that responded affirmatively to a minimum of eight questions were selected for further analysis. Concurrent with data extraction, this quality evaluation was undertaken to ensure that the findings were of significant relevance to the systematic review [57].

2.4. Data extraction and synthesis

Data from each study that met the selection criteria were independently extracted by two reviewers using a customized data extraction form. The following data were extracted and recorded from each study:

- 1) Population: if the study included healthy participants or participants with a specific condition/diseases.
- 2) Data collection methods
- 3) Data collection device: devices were recorded as either customised or commercial devices. Commercial devices were identified by brand and model.
- 4) Number of pressure sensors: the total number of pressure sensors used on both left and right foot, or the total number of pressure plates.
- 5) Other types of sensors: sensors used other than pressure sensors were noted where possible.
- 6) Ground-truth methods: the accurate and definitive labels or annotations assigned to the training data.
- 7) ANNs: the type of neural network used in the study.
- 8) Software to implement the neural network algorithms.
- 9) Evaluation methods or metrics: the statistics methods, metrics, and comparative algorithms that evaluate the performance of proposed methods.

Table 2
Quality checklist.

No	Quality question
1	Are the research objectives and aims clearly defined?
2	Is the method for collecting the data specified, and are the methods reliable?
3	Was the sample size justified?
4	Was the outcome variable (e.g., the plantar pressure measurement) clearly defined and measured?
5	Is the experiment applied to the dataset appropriately?
6	Are the ANN techniques and other methods used in the study well-defined and deliberate?
7	Could the study be replicated based on the provided information?
8	Is the accuracy of models or methods measured and reported?
9	Were validation techniques, including external validation and cross-validation, used to assess the model's performance?
10	Are the models compared with other methods (e.g., DL models vs. ML, DL vs. statistical methods)?
11	Are the findings and results of the study clearly stated, supported by evidence, and analysed for limitations or biases?

3. Results

A total of 780 studies were initially identified for screening. After removing duplicates, 701 records were screened for eligibility. Of these, 546 records were excluded based on their title and abstract, leaving 155 articles for full-text review. Subsequently, 88 articles were excluded for not meeting the selection criteria. After the quality assessment, 60 studies were left for data extraction and analysis. The PRISMA flow diagram is presented in Fig. 2.

3.1. Population

The systematic review encompassed a wide range of pathologies. 38.3 % of the studies (n = 23) were based on healthy individuals. Frequently studied conditions included Parkinson's disease (PD) (n = 10 studies) and foot deformations (n = 8 studies). Other studies focused on detecting risk of falls (n = 7), work-related musculoskeletal disorders (WMSDs) (n = 3), amyotrophic lateral sclerosis (ALS) (n = 4), among others (Fig. 3).

The total number of healthy participants across all studies was 4682. The average number of healthy participants per study was approximately 74. The number of healthy participants per study ranged from a minimum of 2 to a maximum of 1424. The total number of patients across all studies was 1521. The average number of patients per study was approximately 24. Similar to the distribution of healthy participants, the number of patients per study varied greatly, ranging from 3 to 222. The distributions of participants are displayed in Fig. 4.

The studies were often conducted on small groups of healthy participants, which limits the generalizability of the findings to larger patient populations. For example, 56 studies included only healthy participants (and <100 participants) and only 25 studies recruited patients. Eight studies did not recruit any participants, instead sourced online datasets shared by other researchers from PhysioNet [52,53] and CAD WALK [46]. Despite this limitation, studies evaluating patients with PD, foot deformations, and fall detection have used plantar pressure distribution to understand important gait patterns and specific foot pathologies. This suggests that plantar pressure distribution may be a useful tool for studying lower limb and foot conditions [6] and inform the development of targeted interventions [8].

The studies had various objectives, with the most common being gait analysis, occurring in 22 studies. Other frequent objectives included the analysis of plantar pressure images and GRF, present in 16 and 9 studies respectively. Less frequent objectives included posture analysis, COP analysis, shoe comfort analysis, and ankle angle analysis.

3.2. Data collection methods

There was a wide range of methods adopted for data acquisition across the studies, and precise numerical values or units were not consistently reported. 35 studies reported the time taken to collect the data (range 4 s–10 min). 12 studies measured distance (range 1 m–10 m) instead of time elapsed. Efforts to minimize potential errors were evident in 31 studies, as they reported the implementation of multiple sampling instances.

Regarding the acquisition of plantar pressure data, 22 studies used force plates installed within the flooring infrastructure and 45 studies used insole devices. 25 studies opted to design and utilize their own insoles as data loggers, highlighting a trend toward personalized device design, while the remaining studies relied on commercially available products. With respect to the customized devices, the number of pressure sensors per foot ranged from 4 to 128 (mean = 10.2). 35 studies reported the sampling frequency of their devices, ranging from 50 Hz to 300 Hz (mean = 115.1 Hz). The Footscan system, with its high number of pressure points (1908–4096 pressure points), was the most common system used.

Despite the capacity of high spatial resolution data loggers to furnish

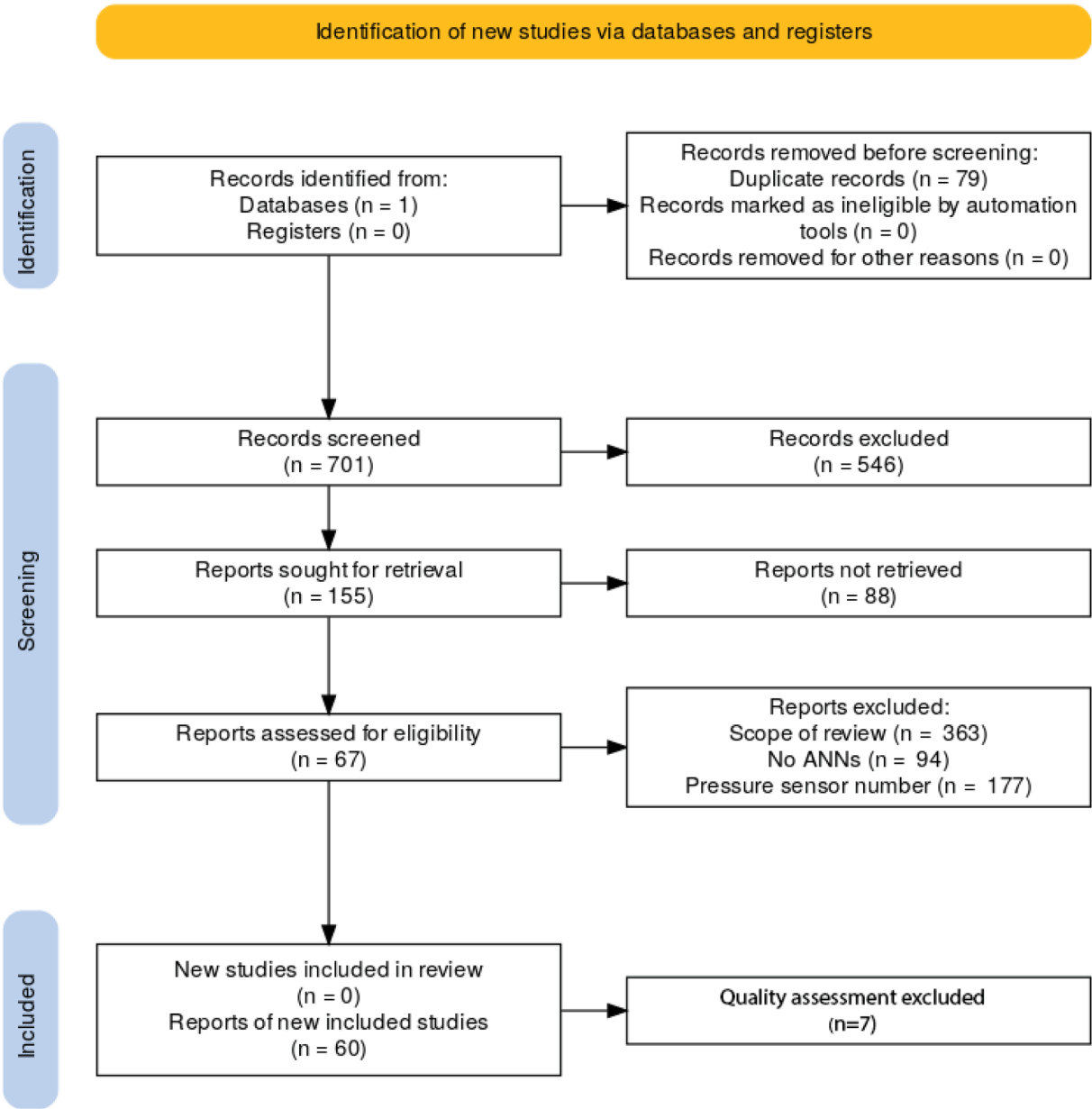


Fig. 2. Diagram of the article-selection process [50].

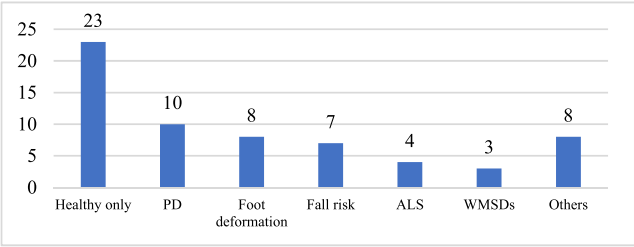


Fig. 3. Distribution of pathologies.

comprehensive details concerning foot conditions, there exists an intriguing paradox in the data processing and analysis stages. For simplification purposes, data obtained from high-resolution sensor array are frequently subdivided into separate regions of interest based on anatomical landmarks through a process of averaging pressure across specific areas. Inevitably, this approach results in a loss of nuanced

shape information pertaining to plantar pressure.

3.3. Multisensory fusion

Inertial measurement units (IMU) were used in combination with pressure sensors in 15 studies and three-dimensional accelerometers were used in combination with pressure sensors in 12 separate studies. (Table 3, Figs. 5 and 6). Three-dimensional gyroscope and magnetometer were used for nine and six studies respectively. Most of these studies (n = 13) focused on gait analysis, while four studies aimed at identifying human posture. Specific foot patterns (e.g., conditions like pes cavus) were the objective of two studies. The use of IMUs reflect the importance of collecting three-dimensional data when analysing plantar pressure as an indicator of lower limb and foot function. Additionally, depth camera data were used in two studies, primarily for converting human video to skeletal structure.

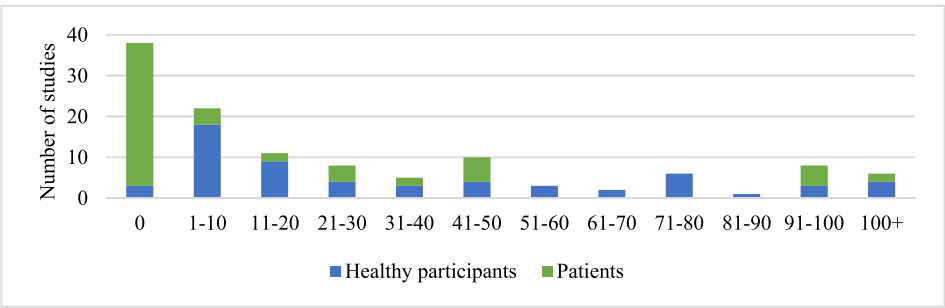


Fig. 4. Stacked distribution of participants.

Table 3
Multisensory fusion.

Article	Objective	Support sensors	Fusion
[60]	GRF	Six-axis force plate	Data-level
[61]	Gait	Four accelerometers	Feature-level
[32]	Gait	Four accelerometers	Feature-level
[22]	Gait	Four accelerometers	Feature-level
[41]	GRF	Depth camera	Feature-level
[46]	Posture	Two accelerometers	Feature-level
[27]	Foot pattern	Two IMU with accelerometers and gyroscopes	Data-level
[62]	Gait	Two IMU with accelerometers and gyroscopes	Data-level
[24]	Gait	Two IMU with accelerometers, gyroscopes, and magnetometers	Data-level
[63]	Gait	Two accelerometers	Data-level
[64]	Posture	Two accelerometers	Feature-level
[65]	Gait	Two IMU with accelerometers and gyroscopes	Data-level
[26]	Musculoskeletal model	Two IMU with accelerometers, gyroscopes, and magnetometers	Data-level
[66]	Gait	Eight electromyography sensors	Data-level
[9]	Foot pattern	Three Video cameras	Feature-level
[67]	Gait	Two IMU with accelerometers, gyroscopes, and magnetometers	Feature-level
[29]	Gait	Depth camera	Feature-level
[42]	Gait	Three-axis piezo-resistive force sensors	Data-level
[39]	Posture	Two IMU with accelerometers and gyroscopes, video camera	Data-level
[34]	Gait	Two IMU with accelerometers, gyroscopes, and magnetometers	Data-level
[68]	Gait	Two IMU with accelerometers, gyroscopes, and magnetometers	Feature-level

3.4. Ground truth

In assessing the validity of the datasets acquired, all studies that collected data from participants ($n = 52$) implemented at least one method of validation wherein the “ground truth” data were reported. As shown in Fig. 7, the most common source of ground truth data is now the “Annotation group”, which is used in 28 studies. The “Annotation group” includes studies where the ground truth was determined by a control group, expert annotation, or medical examination, while 24 studies utilized a reference system to validate the outcomes. Concerning reference devices, digital video cameras, motion capture systems, plantar force plates, and instrumented treadmills were reported in the included studies.

3.5. ANN algorithms

All studies included in this review employed at least one ANN as their main classification tool. CNNs and their variants were proposed by 29 studies (Fig. 7). Recognizing the benefits of time series analysis, RNNs and their variances were implemented in 17 studies. MLPs and FNNs accounted for 17 studies in total. The use of different types of ANN demonstrates the diversity of methods employed to handle different data types and research questions. 13 studies amalgamated more than one ANN algorithms. Moreover, the emergent trend of generative AI has been exploited for sensor placement optimization [18,63]. Transfer learning [6] and deep reinforce learning (DRL) [18,24] were also applied in studies. Fig. 8 and Table 4 lists the ANNs used in the reviewed studies.

It is evident that CNNs and MLPs can be used in applications ranging from static plantar image processing to time-domain gait analysis because of their versatility and robust characteristics in handling different types of data. Simultaneously, applications in that leverage RNNs were used for time-domain signal analysis. This specialization is particularly noticeable in the study of human gait and the investigation of ground reaction forces. The structure of RNNs, which inherently accounts for temporal dependencies, makes them particularly suited to these fields of study.

There is an emerging trend of applying generative AI for sensor

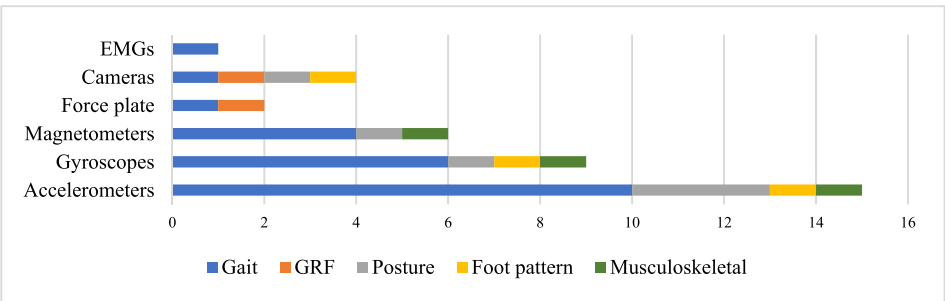


Fig. 5. Sensors to objectives.

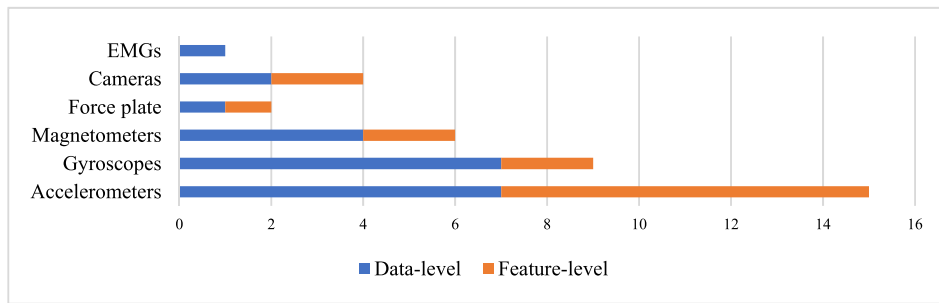


Fig. 6. Sensors to fusion.

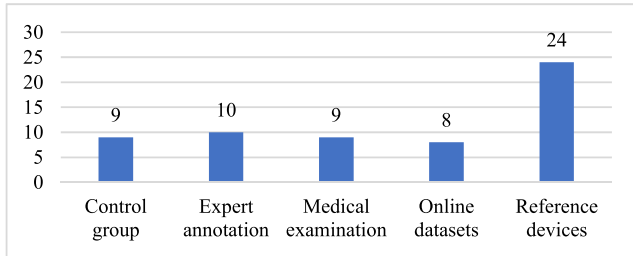


Fig. 7. Number of different ground truth methods.

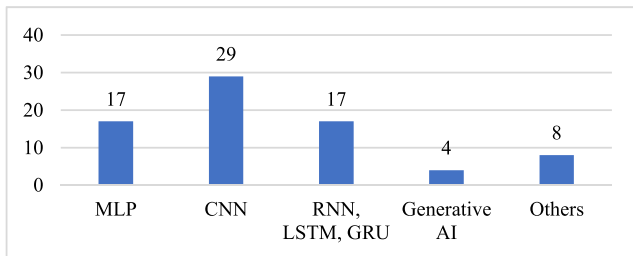


Fig. 8. Number of ANN methods.

placement optimization and data exploration in plantar pressure images. At the time of this systematic review, no studies were identified that apply transformer models in the analysis of plantar pressure distribution.

The studies included in this review predominately used software by MATLAB and Python. Python libraries such as Pytorch, Keras, and Tensorflow were mentioned in these studies.

3.6. Evaluation

The most common statistical approaches were descriptive statistics, correlation coefficients, mean errors, mean square error (MSE) [40], or p-values [20].

Accuracy, precision, sensitivity (recall), specificity, and F1-score [32], area under the receiver operating characteristic curve (AUC-ROC) [68] are commonly used evaluation metrics for classification tasks in machine learning. They are quantitative measures that provide specific numerical values to evaluate the performance of a model. On the other hand, evaluation methods, such as k-fold cross-validation [60] or leave-one-subject-out cross-validation (LOSO-CV) [29] determine how the data are split and how the model's performance is assessed. Evaluation methods aim to provide a comprehensive and reliable assessment of the model's performance by reducing bias, variance, and overfitting. More detailed evaluation methods are presented in Fig. 9.

Various NANN algorithms, such as KNN, NB, SVM, RF, and DT, have been employed as reference models to assess the efficacy of the proposed

ANN algorithms (Fig. 10).

4. Discussion

4.1. Key findings

This systematic review evaluated relevant studies, focusing on the classification of populations, data collection methods, multisensory fusion, ANN algorithms, and evaluation techniques, with several key findings:

- 1) Population classification in experimentation: A notable trend observed was the predominant use of healthy participants in most studies, with the number of patient participants being relatively lower. Furthermore, a common practice among these studies was the utilization of shared online datasets involving the same patient groups. This pattern underscores the challenges inherent in gathering patient-specific data for such research.
- 2) Pressure data collection devices: The review identified a dichotomy in data logging devices, comprising both commercial and custom-designed footwear. Commercial devices demonstrated robustness in terms of accuracy, resolution, and reliability, often serving dual roles in data collection and as benchmarking references. Commercial devices are often equipped with robust functionalities like interest region segmentation, COP, GRF, and in some cases, capabilities as diagnostic assistant tools, and are primarily tailored for clinical or laboratory settings. However, prohibitive cost, challenges in deployment, and substantial power consumption can render them impractical for daily patient monitoring. Furthermore, the data output from these devices necessitates integration with data from other sources for comprehensive analysis, adding an additional layer of complexity for researchers. **In contrast, customized footwear offered greater design flexibility, reflecting an emerging trend towards personalized wearable technology.**
- 3) Utilization of measurement devices: IMUs emerged as a widely accepted motion capture system to supplement plantar pressure analysis [96]. However, the use of other measurement devices such as 3D foot scanners, EMGs, and video cameras remained relatively rare in this domain.
- 4) ANN algorithms in focus: The predominant ANN algorithms identified were MLPs, CNNs, and RNNs. MLPs were favoured for their straightforward architecture. RNNs, known for their efficacy in time-sequence processing, were primarily employed for gait and dynamic plantar pressure analysis. CNNs, with their versatile nature, were suitable for a broad range of analyses.
- 5) Evaluation methods and ground truth assessment: The review summarized various ground truth and evaluation methodologies employed in the context of ANN-based analysis of plantar pressure.

In comparison to other systematic reviews in this field, this study uniquely centres on the application of ANN analysis in plantar pressure research. It comprehensively outlines supportive measurement

Table 4
Pathologies, participants, devices, ground truth, and ANNs techniques used in the included studies.

Article	Pathologies	Healthy	Patient	Device	Ground truth	ANNs
[70]	AIS	10	10	Pedar Insole	Reference devices	MLP
[60]	Healthy only	80	0	Pedar Insole	Reference devices	MLP
[61]	Fall risk	76	24	Footscan insole	Medical examination	MLP
[32]	Fall risk	47	28	Footscan insole	Medical examination	MLP
[22]	ALS	47	28	Footscan insole	Medical examination	MLP
[41]	Healthy only	99	0	Force plate by Kistler	Reference devices	ANN
[71]	WMSDs	10	0	OpenGo	Reference devices	MLP
[46]	Fall risk	3	0	OpenGo	Reference devices	MLP
[59]	PD	73	93	Customized	Online datasets	CNN, LSTM
[58]	PD, ALS, HD	16	48	Customized	Online datasets	CNN, LSTM
[20]	Healthy only	30	0	ZebriS FDM-S system	Reference devices	CNN
[16]	Healthy only	20	10	Footscan force plate	Control group	CNN
[72]	Healthy only	10	0	OpenGo	Reference devices	MLP
[27]	Foot deformation	44	20	Customized	Expert annotation	CNN
[62]	Healthy only	14	0	FootLogger	Expert annotation	CNN
[73]	Fall risk	39	46	Footscan force plate	Medical examination	CNN + LSTM
[24]	Healthy only	10	0	Customized	Reference devices	MLP
[63]	Healthy only	10	0	Customized	Reference devices	CNN
[74]	Foot deformation	2	4	Customized	Control group	MLP
[75]	Foot deformation	96	0	G-hiwell's Low Pile Pressure Measuring Device	Medical examination	CNN
[6]	Foot deformation	151	0	FreeMed platforms by Piédica	Control group	MLP
[64]	PD, ALS, HD	2	0	OpenGo	Reference devices	MLP
[76]	PD	73	93	Customized	Online datasets	Fuzzy neural network
[77]	PD	73	93	Customized	Online datasets	ANN
[78]	PD, ALS, HD	16	48	Customized	Online datasets	MLP
[65]	Foot deformation	44	36	Customized	Control group	CNN
[79]	Healthy only	60	0	RSScan	Reference devices	CNN
[23]	Healthy only	15	0	Customized	Reference devices	DRL
[80]	Foot deformation	0	222	Customized	Control group	ANN
[81]	PD	5	0	Footscan insole	Control group	CNN
[26]	Healthy only	10	0	Wearable Gait Lab	Reference devices	DRL
[66]	Healthy only	3	0	Customized	Reference devices	LSTM, MLP
[82]	Healthy only	7	0	Customized	Reference devices	CNN
[9]	Healthy only	100	0	Customized	Expert annotation	CNN
[83]	Healthy only	6	0	Footscan insole	Reference devices	CNN
[47]	Fall risk	22	8	Footscan insole	Medical examination	MLP, Autoencoder
[84]	PD	73	93	Customized	Online datasets	CNN
[69]	Healthy only	6	0	Footscan force plate	Reference devices	Genetic Algorithm
[85]	Foot deformation	60	0	Footscan force plate	Expert annotation	CNN, GAN
[86]	Healthy only	12	0	Footscan force plate	Control group	ANN
[87]	Healthy only	8	0	Footscan force plate	Expert annotation	CNN
[58]	Multiple sclerosis	62	43	Win-Track platform	Control group	CNN
[67]	Healthy only	11	0	Customized	Expert annotation	CNN, LSTM, MLP
[29]	Gait abnormal	20	0	pressure plate by GHiWell	Expert annotation	CNN, LSTM
[88]	PD	0	11	Footscan insole	Medical examination	LSTM
[46]	Foot deformation	55	50	Footscan force plate	Online datasets	GAN
[89]	Multiple sclerosis	62	43	Win-Track platform	Medical examination	CNN
[90]	DM	265	207	eMat foot pressure distribution test system	Control group	Improved Capsule Network
[91]	Knee pain and knee osteoarthritis	1424	0	Unknown	Expert annotation	CNN
[92]	Cerebral palsy	974	0	Customized	Expert annotation	CNN
[10]	Healthy only	18	0	Pedar Insole	Reference devices	RNN
[93]	Fall risk	25	23	Customized	Medical examination	CNN, LSTM
[64]	WMSDs	10	0	OpenGo	Reference devices	LSTM, GRU
[40]	Healthy only	40	0	Customized	Reference devices	LSTM
[42]	Healthy only	9	0	Customized	Reference devices	CNN, RNN, MLP
[94]	Healthy only	30	0	Customized	Reference devices	LSTM
[95]	PD	73	93	Customized	Online datasets	LSTM
[39]	Healthy only	8	0	Customized	Expert annotation	LSTM
[34]	Fall risk	32	0	Customized	Reference devices	CNN + RNN
[68]	PD	0	32	Footscan insole	Reference devices	CNN, LSTM

In ANNs column, algorithms separated by comma means the study utilised different approaches for same purpose, while the “+” means the approach is the combination and variance of two algorithms.

Adolescent Idiopathic Scoliosis (AIS), Huntington’s disease (HD).

methodologies and experiment setups, as well as data analysis techniques that leverage ANNs for insightful plantar pressure analysis.

4.2. Implications of foot plantar pressure analysis via ANNs

Clinical and biomechanical repercussions: The integration of ANN methodologies with foot plantar pressure analysis heralds significant advancements in the clinical and biomechanical fields. This synergy

offers a potent diagnostic, monitoring, and predictive tool for an array of medical conditions. Particularly noteworthy is its application in the analysis of gait-related disorders, such as PD [97] and fall risk detection [73], where ANNs demonstrate an acute sensitivity in identifying and categorizing these conditions. Additionally, the capability of ANNs in discerning and classifying foot deformities presents a substantial leap forward in both diagnostic and therapeutic domains [98].

Implications for footwear device industry: For manufacturers and

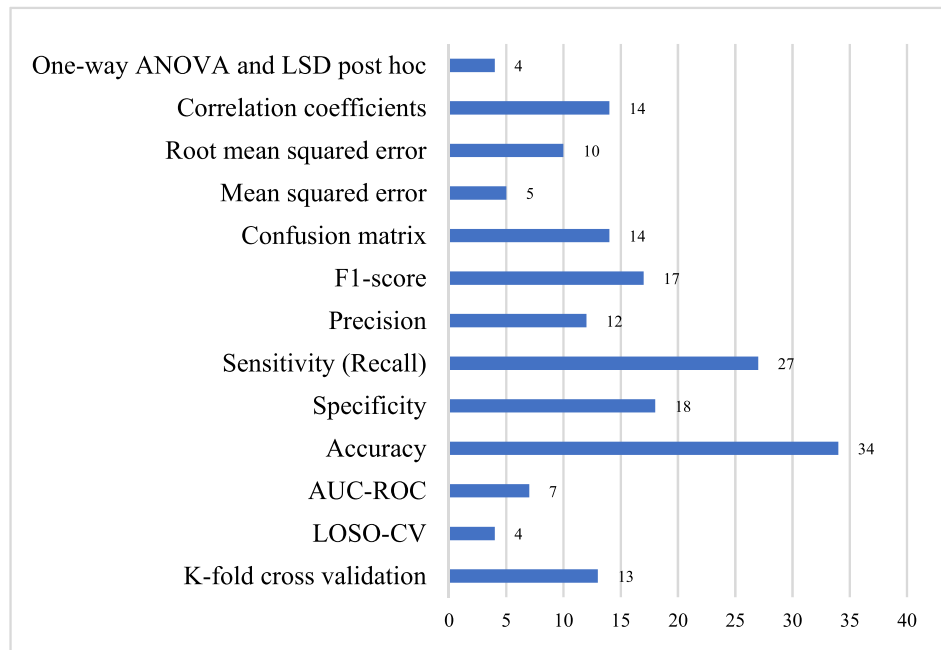


Fig. 9. Evaluation methods and matrices.

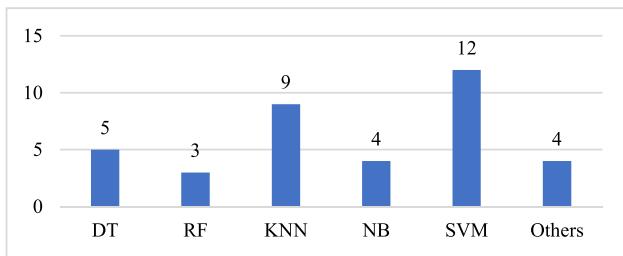


Fig. 10. NANN methods as evaluation benchmarks.

suppliers of foot orthosis devices, the application of ANNs in plantar pressure analysis is of paramount importance. This technology is instrumental in addressing critical aspects such as pressure off-loading [99–102], design and optimization of device shape [103].

4.3. Unresolved questions and challenges

Despite the encouraging progress made in utilizing ANN techniques for foot plantar pressure analysis, the field confronts several unresolved questions and challenges that open new avenues for research:

- 1) **Multimodal data integration in plantar pressure analysis:** A pivotal area of exploration lies in the integration of plantar pressure data with other biomechanical measurement methods. The intrinsic relationship between plantar pressure distribution and supplementary diagnostic techniques, such as IMU based gait motion capture systems and 3D foot plantar scans [104], warrants detailed investigation. Understanding how to effectively amalgamate and interpret these disparate data sources could significantly enhance the comprehensiveness and accuracy of foot biomechanics analysis, thereby fostering advances in measurement fusion techniques [105].
- 2) **The future of employing ANNs in the management of DFUs** is promising, offering a paradigm shift in early detection and intervention. ANNs hold the potential to analyse complex patterns in plantar pressure and foot temperature, in indicators of DFU risks [13]. This advanced technology could enable more precise and personalized healthcare, allowing for earlier identification of at-risk

patients and more effective treatment strategies. However, the successful integration of ANNs in clinical practice necessitates overcoming challenges related to data accuracy, algorithmic reliability and ensuring equitable healthcare outcomes across diverse patient groups [106]. Embracing these advancements could significantly enhance DFU management, leading to improved patient care and outcomes.

- 3) **Long-term monitoring with wearable devices:** Investigating the feasibility and efficacy of wearable devices for long-term plantar pressure monitoring presents a critical research frontier. These devices hold immense potential for transforming foot health monitoring, particularly in managing chronic conditions and preventive healthcare. However, the practical implementation of such devices, alongside considerations for user compliance and the accuracy of data in everyday use, necessitates exhaustive evaluation [107].
- 4) **Advancements in deep learning techniques:** The field of deep learning is rapidly evolving, marked by significant developments such as CNNs for image recognition [43] and LSTM networks for time sequence analysis [44], which have already enhanced plantar pressure analysis. The potential incorporation of cutting-edge models like Transformers and attention mechanisms, which promise to outperform traditional methods, deserves thorough exploration [48]. Assessing their effectiveness relative to established machine learning and statistical approaches is crucial for advancing the field.
- 5) **Efficiency and cost-effectiveness of transfer learning in ANN Models:** Customized ANN models for plantar pressure evaluation, while beneficial, are resource-intensive and time-consuming to train, especially due to the challenges in sourcing patient-specific data. A more efficient alternative may lie in employing pretrained AI models through transfer learning [108,109]. This approach can harness the power of existing models trained on extensive datasets, adapting them for plantar pressure analysis. Such a strategy could offer a cost-effective and efficient solution, mitigating the limitations posed by the scarcity of patient data for researchers and clinicians [110].

These areas of inquiry underscore the dynamic and evolving landscape of ANN application in foot plantar pressure analysis. By addressing these challenges, the field can not only deepen its understanding but also pave the way for innovative solutions in foot health and biomechanics,

signifying a fertile ground for future research endeavours.

5. Conclusion

This systematic review comprehensively analyses the use of ANN techniques in foot plantar pressure analysis, highlighting the versatility of ANN techniques in assessing different pathologies, capability to integrate data from various devices, and potential for wearable devices. Distinguishing itself through a methodical classification approach, the review evaluates populations, data collection methods, multisensory fusion, ANN algorithms, and evaluation techniques, revealing key insights such as the predominance of healthy participants in studies, the balance between commercial and custom-designed footwear, and the effectiveness of IMUs and other sensors.

The review underscores the necessity of integrating plantar pressure data with other measurement methods, exploring wearable devices for long-term monitoring, and keeping abreast of advancements in deep learning. This focus on ANN applications in plantar pressure studies sets the review apart in biomechanical research and paves the way for future explorations that could lead to innovative solutions in foot health and biomechanics.

CRediT authorship contribution statement

Chongguang Wang: Data curation, Writing – original draft, Visualization, Investigation, Validation. **Kerrie Evans:** Writing – review & editing, Methodology, Supervision, Project administration. **Dean Hartley:** Supervision, Resources, Funding acquisition. **Scott Morrison:** Formal analysis, Methodology, Supervision, Resources. **Martin Veidt:** Supervision, Resources. **Gui Wang:** Conceptualization, Funding acquisition, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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